

SYSTEM REQUIREMENTS SPECIFICATION

Backend Architecture Plan • AI-Powered Job Tracking SaaS

Core Architectural Assumption: Multi-Tenant Workspace Isolation. The system is a private workflow tracker. It does not actively scrape external job boards. Users input data manually or via text injection.

1. Product Vision & MVP Scope

The objective is to build a robust, scalable backend for a multi-tenant job application tracking platform. Data isolation is maintained strictly at the workspace level rather than the individual user level, enabling multi-user collaboration within separate tracking contexts.

Minimum Viable Product (MVP) Core Features

- **Identity Management:** User registration, secure login, and password resetting.
- **Workspace Multi-Tenancy:** Creation of workspaces and email-based membership invitations.
- **Job Tracking:** Manual creation, status updates, and archiving of job applications.
- **Task & Context Logging:** Plain text or markdown notes alongside deadline-driven reminders.
- **AI Contextualization:** Asynchronous text processing to extract details and summarize job listings.
- **Aggregated Statistics:** Basic metric calculation for the user landing dashboard.

2. Membership & Access Control Roles

- **Owner:** Full workspace authorization, including subscription management, settings adjustment, and workspace deletion rights.
- **Admin:** Complete operational management of jobs, notes, and member permissions, excluding workspace deletion.
- **Member:** Full access to create, edit, view, and process job entries within authorized workspaces.
- **Viewer:** Read-only privileges across all applications, metrics, and logs inside the designated workspace.

3. Structural Domain Model & Database Schema

Every single query executed against application-scoped data must programmatically validate and filter by `workspace_id` to prevent cross-tenant data leakages.

Table: users	
Primary Key	id (UUID / String)
Attributes	email, password_hash, name, avatar_url (optional), is_active (boolean)
Timestamps	created_at, updated_at, last_login_at

Table: workspaces	
Primary Key	id (UUID / String)
Attributes	name, slug, owner_user_id (FK to users), plan (optional), timezone
Timestamps	created_at, updated_at, archived_at (optional)

Table: workspace_members	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), user_id (FK)
Attributes	role (owner, admin, member, viewer), status (active, invited, removed), invited_by_user_id (FK)
Timestamps	joined_at, created_at

Table: workspace_invitations	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), invited_by_user_id (FK)
Attributes	email, role, token, status (pending, accepted, revoked, expired)
Timestamps	expires_at, created_at, accepted_at

Table: job_applications	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), created_by_user_id (FK)
Attributes	title, company_name, source, source_url, location, employment_type, seniority_level, status, priority, salary_min, salary_max, currency
Timestamps	applied_at, deadline_at, archived_at, created_at, updated_at, deleted_at (Soft Delete)

Table: job_status_history	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), job_application_id (FK), changed_by_user_id (FK)
Attributes	from_status, to_status, comment (optional)
Timestamps	changed_at

Table: job_notes	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), job_application_id (FK), author_user_id (FK)
Attributes	content (text / markdown supported)
Timestamps	created_at, updated_at, deleted_at (optional)

Table: reminders	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), job_application_id (FK), created_by_user_id (FK)
Attributes	title, due_at, status (pending, sent, done, cancelled), channel (email, in_app), recurrence_rule (optional)
Timestamps	created_at, updated_at

Table: ai_job_insights	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), job_application_id (FK)
Attributes	type (summary, match_score, follow_up_email), input_hash, output_text, model_name, status (pending, completed, failed)
Timestamps	created_at, completed_at

Table: audit_logs	
Primary Key	id (UUID / String)
Foreign Keys	workspace_id (FK), actor_user_id (FK)
Attributes	entity_type, entity_id, action, before_json, after_json
Timestamps	created_at

4. Application Lifecycle & Pipeline Model

Job applications progress through discrete pipelines chronologically represented via the following lifecycle model transition:

Saved → Applied → Screening → Interview → Offer → [Rejected / Withdrawn]

5. Non-Functional & Execution Guidelines

- **Security & Isolation:** Mandatory verification of workspace membership on every incoming request. Strict token-based session verification with protected rate limits applied onto high-cost LLM/AI interaction routes.
- **Scalability & Performance:** Enforce cursor or offset pagination across analytical and query lists. Dashboard aggregation queries should be cached where practical.
- **Asynchronous Processing:** Long-running AI operations and notification dispatches must run outside the request-response thread using reliable background message workers.
- **Implementation Order Strategy:** Build in distinct incremental stages. Establish foundational infrastructure (*users* → *workspaces* → *job_applications*) prior to engineering notification layers, operational log engines, or external AI API integration layers.